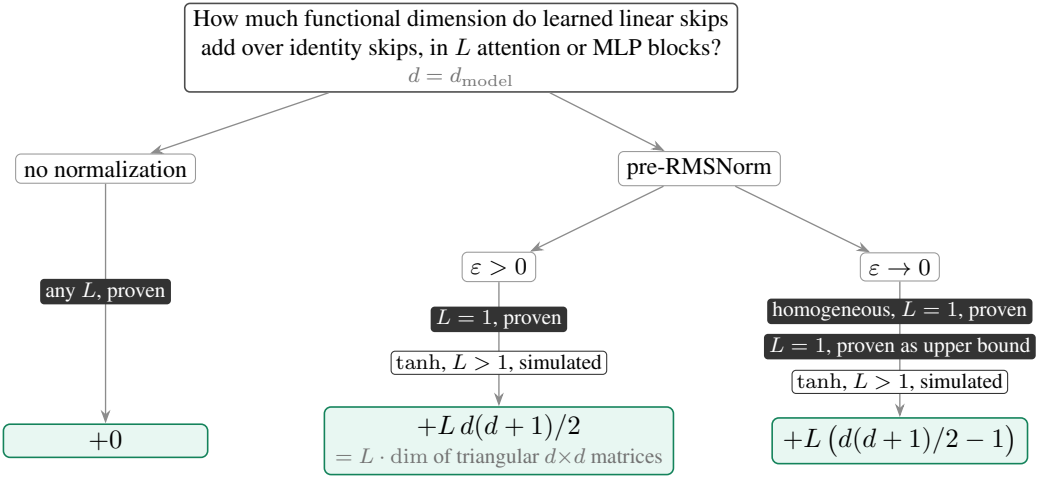

How much functional dimension do learned linear skips add over identity skips, in L attention or MLP blocks?
$d = d_{\text{model}}$
no normalization
pre-RMSNorm
any L, proven
$+0$
$\varepsilon > 0$
$L = 1$, proven
tanh, $L > 1$, simulated
$+L\, d(d+1)/2$
$= L \cdot$ dim of triangular $d \times d$ matrices
$\varepsilon \to 0$
homogeneous, $L = 1$, proven
$L = 1$, proven as upper bound
tanh, $L > 1$, simulated
$+L\left(d(d+1)/2 - 1\right)$